Paul Pokliatskyi | UMass Boston | Feb 2026 | https://github.com/WombaWingu/Volatility-Forecast

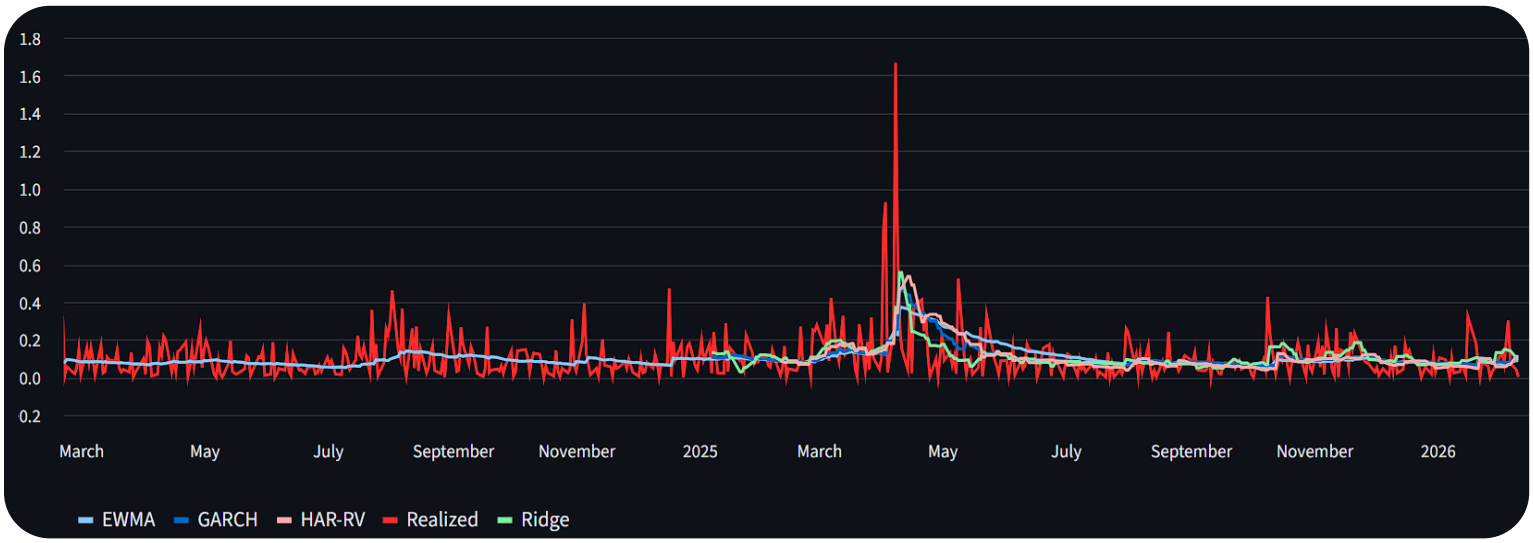
Project Description

A Python equity volatility-forecasting research pipeline that trains/compares multiple models, evaluates them (stats and risk backtests), and returns plots/metrics as artifacts (dashboard + cross-sectional extensions available in the repo)

What I implemented

- End-to-end Python pipeline for rolling out-of-sample volatility forecasting on SPY(daily)
- Models: EWMA, HAR-RV, GARCH, Student-t GARCH, GJR, and Ridge (+ simple baselines)
- Outputs: saved plots + metrics tables as artifacts (reproducible runs)
- **S&P500 Volatility Forecast Plot + Metrics**

SPY Realized volatility from daily returns; 1-day-ahead volatility forecasts (01/01/2020 – 02/23/2026)



Headline Metrics
Lower is Better

Model	MAE	RMSE	MSE_var	Log_vol
naive	0.1123	0.1694	0.0306	4.0549
roll_mean	0.0922	0.1440	0.0253	2.3908
har_rv	0.0866	0.1501	0.0341	2.0283
ewma	0.0892	0.1405	0.0247	2.3252
garch	0.0818	0.1451	0.0322	1.9411
gjr	0.0793	0.1399	0.0316	1.9090
garch_t	0.0819	0.1448	0.0321	1.9409
ridge	0.0759	0.1339	0.0306	1.8295

Performance by Regime
(Realized volatility terciles (Low/Medium/High))

Regime	MAE	RMSE	QLIKE
High	0.1519	0.2451	3.4584
Low	0.0722	0.0858	2.5465
Medium	0.0324	0.041	0.3569